

Matthew McCall

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Skills: C++, GNU/Linux, Git, MATLAB, KiCad, SPICE, SystemVerilog, SolidWorks, LaTeX, MS Word, Excel, PowerPoint

Education

Rensselaer Polytechnic Institute	Aug 2022 – (Expected) May 2026
▪ Bachelor's in Computer Science and Computer Systems Engineering, Class of 2026	
Benjamin N. Cardozo High School	Sep 2018 – Jun 2022
▪ High School Diploma with Regents Advanced Designation with Mastery in Mathematics, Class of 2022	
▪ Member of the National Honors Society (ARISTA)	

Work Experience

Engineering and Quality Intern at HarcoSemco	Sep 2024 – Dec 2024, May 2025 – Aug 2025
▪ Led the end-to-end development of a Bluetooth-based asset tracking system within an aerospace environment. Designed the network architecture of gateways and tags, selected hardware, developed C++, Java, and Kotlin software, and deployed the system to track assets across a 12,000 sq. ft area with 8-foot accuracy.	
▪ Engineered C++ software for detecting potential toxic gases using Bosch sensors in a regulated aerospace setting, transmitting data via Bluetooth. Also developed a React TypeScript dashboard using Web Bluetooth to display sensor data in real time, ensuring all components met aerospace-grade compliance and safety protocols.	
Undergraduate Research Assistant at SCOREC	Sep 2022 – Aug 2024, Jan 2025 – Present
▪ Reduced runtime by 30% by implementing a GPU memory pool in Omega_h, a C++ library for triangle mesh adaptivity.	
▪ Implemented an algorithm in C++ for finding the nearest simplex to a point in an n-dimensional mesh.	
▪ Authored a scientific paper and poster in LaTeX. I presented my work at two poster sessions at RPI.	
Mercer X Lab Assistant	May 2024 – Aug 2024, Jan 2025 – Present
▪ Guided students through safety best practices, circuit design, soldering, equipment usage.	

Project Experience

Open Algebra Software for Inferring Solutions (OASIS)	Sep 2023 – Present
▪ Currently leading the team for the OASIS project, an open-source C++ library for computer algebra and symbolic manipulation. I used GitHub Issues, Projects, and Pull Requests to manage the project. I also implemented GitHub Actions CI/CD workflows for automated building and unit testing of the project.	
Eye Robot	May 2024 – Aug 2024
▪ Eye Robot is a robot that assists the visually impaired in navigation. I worked in a team to write C# and C++ software to construct a 3D model of the room using LiDAR and send commands to an onboard ESP32.	
HackRPI 2025	November 2025
▪ Won Best AI Hack and Best Use of DigitalOcean Gradient AI by creating a mock interview platform that uses AI agents to simulate an interviewer persona and give feedback.	

Volunteer Experience

Rensselaer Center for Open Source (RCOS) Mentor	May 2024 – Aug 2024, Jan 2025 – May 2025
▪ Served as an RCOS mentor, providing guidance and technical support for multiple projects.	
Volunteer at The Forge, An RPI Student Makerspace	May 2024 – Aug 2024, Jan 2025 – May 2025
▪ Assisted students with computer aided design and 3D printing	
Live Stream Assistant at St Mary Church, Branford CT	Jun 2022 – Present
▪ Routinely aids with executing livestreams of Sunday Mass.	